HyperAPI Hackathon

Stress-test HyperAPI (Alpha) against the hardest document intelligence problems in Finance & Accounting.

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The Financial Gauntlet

Mar 14, 2026 — 12:00 PM – 6:00 PM

JBR Techpark, Whitefield

Powered by **Amazon Web Services** as the official infrastructure partner, with support from **AWS for Startups**.

AWS Cloud Resources

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☆ The Challenge

You are given **gauntlet.pdf** — a 1,000-page Accounts Payable processing bundle for a fictional company, **HyperAPI Technologies Pvt Ltd**, covering FY 2025–2026.

Hidden inside are **200 deliberate errors ("needles")** across 20 categories. Your job: build an AI pipeline that finds as many as possible, as accurately as possible.

Download gauntlet.pdf (3.3 MB)

What's in the PDF

PAGES	CONTENT
1	Cover page
2	Submission format & scoring rules
3–4	**Vendor Master** — the authoritative list of 35 registered vendors (names, GSTINs, bank IFSCs, states). Cross-reference everything against this.
5–1000	~750 documents: tax invoices, purchase orders, bank statements, expense reports, credit/debit notes, receipts, quotations, delivery notes, T&Cs, and filler

Needle Categories (20 types, 3 difficulty tiers)

Easy 40 needles × 1 pt = 40 pts

CATEGORY	COUNT	WHAT TO LOOK FOR
`arithmetic_error`	12	qty × rate ≠ amount, subtotal ≠ sum of lines, tax calc wrong, or grand total ≠ subtotal + tax
`billing_typo`	4	Hours logged as 0.15 (decimal) when it means 0:15 = 0.25 hrs. The rate × wrong qty gives wrong amount
`duplicate_line_item`	4	Exact same line item appears twice in one invoice, doubling the charge
`invalid_date`	10	Impossible dates: Feb 31, Sep 31, day 00, day 32, Feb 29 in non-leap year
`wrong_tax_rate`	10	GST rate doesn't match the HSN/SAC code (e.g., freight at 18% instead of 5%)

Medium 60 needles × 3 pts = 180 pts

CATEGORY	COUNT	WHAT TO LOOK FOR
`po_invoice_mismatch`	15	Invoice qty or rate differs from the linked Purchase Order

CATEGORY	COUNT	WHAT TO LOOK FOR
`vendor_name_typo`	10	Vendor name on invoice is misspelled vs the Vendor Master (pages 3–4)
`double_payment`	10	Same payment (vendor, amount, reference) appears in **two different** bank statements months apart
`ifsc_mismatch`	5	Bank IFSC on invoice doesn't match vendor's registered IFSC in Vendor Master
`duplicate_expense`	10	Same expense entry claimed in two different expense reports
`date_cascade`	5	Invoice date is **before** its own PO date (can't invoice before the PO exists)
`gstin_state_mismatch`	5	First 2 digits of GSTIN (state code) don't match the vendor's address state

Evil 100 needles × 7 pts = 700 pts

CATEGORY	COUNT	WHAT TO LOOK FOR
`quantity_accumulation`	35	Sum of quantities across 3–5 invoices exceeds the PO quantity by 20%
`price_escalation`	10	All 4 invoices against a PO charge rates exceeding the contracted PO rate
`balance_drift`	15	Bank statement opening balance ≠ previous month's closing balance
`circular_reference`	8	Credit/debit notes form a loop (A→B→C→A) — none traces back to a real invoice
`triple_expense_claim`	10	Same hotel stay claimed in 3 different expense reports
`employee_id_collision`	7	Same Employee ID used by two different people across expense reports
`fake_vendor`	10	Invoice from a vendor that doesn't exist in the Vendor Master
`phantom_po_reference`	5	Invoice cites a PO number that doesn't exist anywhere in the dataset

Total: 200 needles, 920 max points

Submission Format

Submit a single JSON file:

```
{
  "team_id": "your_team_name",
  "findings": [
    {
      "finding_id": "F-001",
      "category": "arithmetic_error",
      "pages": [47],
      "document_refs": ["INV-2025-0042"],
      "description": "Line item 1 total should be ...",
      "reported_value": "130,123.87",
      "correct_value": "129,623.87"
    }
  ]
}
```

Scoring

What's scored vs not

FIELD	SCORED?	NOTES
`category`	**Yes**	Must match exactly. Used for matching your finding to an answer key needle.
`document_refs`	**Yes**	≥50% overlap with answer key required for a match.
`pages`	**Yes**	+/−1 page tolerance.
`reported_value`	**Yes**	5% numeric tolerance (or exact string match).
`correct_value`	**Yes**	5% numeric tolerance (or exact string match).
`finding_id`	No	Your internal tracking ID.

FIELD	SCORED?	NOTES
`description`	No	Free text for human review. Not scored.

How scoring works

Each finding is first **matched** to an answer key needle (same `category` + ≥50% `document_refs` overlap). Then:

```
needle_score = difficulty_weight × (0.4 × detection + 0.3 × localization + 0.3 × diagnosis)
```

COMPONENT	WEIGHT	HOW IT'S CALCULATED
Detection	0.4	1.0 if matched, 0.0 if not
Localization	0.3	Fraction of expected pages you identified (±1 tolerance)
Diagnosis	0.3	0.5 × (reported_value match) + 0.5 × (correct_value match)

Difficulty weights: easy = 1, medium = 3, evil = 7

False positive penalty: −0.5 per unmatched finding, capped at 20% of your earned score.

Example

You find an evil needle perfectly: `7 × (0.4×1.0 + 0.3×1.0 + 0.3×1.0) = 7.0 pts`

You find it but wrong page and no values: `7 × (0.4×1.0 + 0.3×0.0 + 0.3×0.0) = 2.8 pts`

Tips

1. **Start with the Vendor Master (pages 3–4).** Many needles are cross-reference checks against it.
2. **Easy needles are single-document checks** — arithmetic, dates, tax rates. Good for quick wins.
3. **Medium needles require comparing 2 documents** — PO vs invoice, vendor master vs invoice, two expense reports.
4. **Evil needles require aggregating across 3–5+ documents** scattered across hundreds of pages — cumulative quantities, price trends, circular references.

5. **Don't over-report.** False positives cost you points. Be precise.
6. **Document IDs matter.** `INV-2025-00042` is not the same as `INV-2025-0042`. Copy them exactly.